

Ideation Phase

Brainstorm & Idea Prioritization Template

Metric	Details
Date	08 July 2026
Institution	Vishnu Institute of Technology
Project Name	Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease
Maximum Marks	4 Marks

Ideation Phase - Clinical User Empathy Map

An Empathy Map provides a visual approach to understanding a user's cognitive and emotional experiences[cite: 5]. This document captures the internal and external realities of clinical diagnostic professionals handling vast row-by-row patient datasets[cite: 5]. By mapping out these criteria, we align our software grid architecture to target explicit user needs, stripping out irrelevant features to highlight critical medical indicators[cite: 5].

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The principal developer came together to define the scope of the project[cite: 5]. Relevant background on health informatics datasets and prior Tableau dashboard examples was shared ahead of the session, and all core parameters were confirmed before the development sprint began[cite: 5].

PROBLEM STATEMENT

How might we analyse multi-variable clinical heart disease records and identify cardiovascular risk factor trends using data analytics, so that diagnostic health specialists and clinical care teams can make informed, data-driven preventative treatment decisions about patient management?[cite: 5]

Key Rules of Brainstorming Followed

✓ Stay on topic Keep every idea tied to the problem statement on heart disease.[cite: 5]	✓ Encourage wild ideas Unconventional analytical angles were welcomed before being filtered.[cite: 5]
☑ Defer judgment No idea was rejected during the listing stage; evaluation came later.[cite: 5]	✓ Listen to others Each member reviewed and built upon ideas shared by teammates.[cite: 5]

Step-2: Brainstorm, Idea Listing and Grouping

Individual Idea Listing (Sticky-Note Stage)

Each quadrant independently details the core mindset, environment, targets, and limitations confronting the persona during diagnostics workflows[cite: 5].

What They Think & Feel	What They See	What They Say & Do	Experience Analysis
"I need to spot compounding high-risk factor clusters across my patient list rapidly before critical events take place." [cite: 5]	A massive influx of patient diagnostic spreadsheets containing thousands of rows and multiple fields.[cite: 5]	Says: "Show me exactly how lifestyle inputs like smoking/alcohol correspond to stroke history."[cite: 5]	Pains: Connection timeout disruptions and application sync failures occur.[cite: 5]
"Static medical data spreadsheets are incredibly tedious and time-consuming to read when cross-referencing." [cite: 5]	Cluttered UI templates filled with unnecessary distracting corporate branding elements. [cite: 5]	Does: Selects and filters interactive metrics panels to isolate specific demographic groups. [cite: 5]	Pains: Rigid systems lack simultaneous dual-axis view ports for summaries.[cite: 5]
"I feel overwhelmed by generic corporate software layouts that waste screen space."[cite: 5]	Clean, professional charts where risk trends instantly emerge based on age distributions.[cite: 5]	Utilizes clean visual workspaces to verify risk clusters quickly.[cite: 5]	Gains: A custombuilt, lightweight web dashboard wrapper loads live data smoothly.[cite: 5]
Focuses heavily on identifying critical indicators to reduce cognitive clinical analysis overload.[cite: 5]	Targets explicit parameters to bypass non-essential UI features.[cite: 5]	Eliminates manual lookup blocks to speed up clinical validation.[cite: 5]	Gains: Clear structural organization exists from ideation through testing logs. [cite: 5]

Step-3: Idea Prioritization

With the clustered categories in hand, the team mapped these observations directly against system implementation feasibility guidelines to establish clear design constraints for the web layout[cite: 5].

Prioritization Summary

Quadrant	Target Technical Actions Allocated
Group	
Do First	Resolving text cognitive overload; constructing real-time clinical visual metrics layouts. [cite: 5]
Plan Next	Integrating complex multi-variant predictions (kept as future execution scope).[cite: 5]
Quadrant	

Target Technical Actions Allocated

Group	
Fill-in Tasks	Developing clean card components; embedding cloud frames; structuring Git profiles. [cite: 5]
Reconsider	Deploying stock boilerplate layout structures with corporate filler text strings.[cite: 5]

Conclusion & Project Scope Determination

The ideas placed in the "Do First" quadrant formed the core foundation of the completed platform[cite: 5]. The critical user demand for immediate multi-parameter cross-filtering was answered by building the fixed dark slate parameters index panel, establishing a direct link between user empathy analysis and full-stack software implementation[cite: 5].